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1. Implementation and per-tokenizer training. All three tokenizers share one BPE algorithm: character-level start, iterative frequency-based pair merging, and 32 direct two-word bigram tokens. Tokenizer 1 (domain 1) uses vocab_size=4000; tokenizer 2 and tokenizer 3 use vocab_size=5000. The vocab_size for tokenizer 1 was tuned by sweeping 9 values between 1000 and 8000 and measuring real NER F1 on domain_1_dev.txt: F1 peaked at 0.4931 at vocab_size=4000 (vs. 0.4572 at 5000), with no loss in encoding speed or tokens/char, and is reported as an observed dev-set effect, not a guaranteed law for the hidden test file. The only behavioral change kept in the final code is a training-speed optimization: instead of rescanning the whole corpus to recount pair frequencies after every merge, the algorithm keeps one running pair-count table plus an index from each pair to the sequences that contain it, updating only the sequences a merge actually changes. This produces identical merges, vocabulary, and encoded output to the naive algorithm (verified by direct comparison) while cutting tokenizer training time substantially. An earlier attempt to bias merge frequencies toward capitalized/entity-like words was tested and reverted after lowering domain-1 F1 (0.4487 to 0.4300); the final code uses plain frequency counts only. Tokenizer 3 uses the identical algorithm, trained via generate_tokenizers.py with --train_files_3 domain_1_train.txt domain_2_train.txt, since it must serve a domain unseen at training time. vocab_size=4000 was also tried for tokenizer 3 and rejected: it raised domain-2 F1 only marginally (0.9611 to 0.9646) while lowering domain-1 F1 (0.4695 to 0.4631) and tokens/char efficiency, so tokenizer 3 keeps vocab_size=5000 -- the tokenizer-1 sweet spot did not transfer to the differently-composed combined corpus.

2. Top-5 / bottom-5 most frequent bigram tokens (of the 32 direct two-word tokens selected per tokenizer):

Tokenizer 1 top: MTV Movie (449), SO MUCH (205), I'm SO (116), GOOD MORNING (112), IM SO (97). Bottom: REALLY REALLY (44), WHY DO (42), MOTHER'S DAY (42), MOVIE AWARDS (37), CAN DO (35).

Tokenizer 2 top: Donald Trump (3526), New York (2181), Hillary Clinton (1215), White House (1180), Donald Trump's (878). Bottom: Keystone XL (65), GOP Congressman (57), GOP Debate (55), NEW YORK (51), RELATED LINKS (28).

Tokenizer 3 top: MTV Movie (493), HUFFPOST HILL (245), SO MUCH (205), I'm SO (116), GOOD MORNING (112). Bottom: IM NOT (49), SO SAD (47), SO EXCITED (45), REALLY REALLY (44), MOTHER'S DAY (43).

3. Dev-set F1 and tokenizer efficiency:

Tokenizer 1 (domain 1 on domain 1): vocab 4000, speed 391,980 tok/s, tokens/char 0.3221, reconstruction 100%, NER F1 0.4931.

Tokenizer 2 (domain 2 on domain 2): vocab 5000, speed 212,388 tok/s, tokens/char 0.2976, reconstruction 100%, NER F1 0.9623.

Tokenizer 3 (combined on domain 1): vocab 5000, speed 210,747 tok/s, tokens/char 0.3160, reconstruction 100%, NER F1 0.4695.

Tokenizer 3 (combined on domain 2): vocab 5000, speed 330,894 tok/s, tokens/char 0.3178, reconstruction 100%, NER F1 0.9611.

4. Assessment of tokenizer 3 on the hidden domain. No labeled data exists for the hidden domain, so likely performance is assessed with visible-domain generalization probes rather than a direct measurement. Vocabulary overlap: tokenizer 3 shares 74.4% of its vocabulary with tokenizer 1 and 61.4% with tokenizer 2, far higher than the 31.9% the two domain-specific tokenizers share with each other; tokenizer 3 is a genuine blend, not just one domain copied. Cross-domain compression on random 8,000-line dev samples: tokenizer 3 reaches 1.75 tokens/word and 0.05% UNK on domain 1, and 1.87 tokens/word and 0.04% UNK on domain 2, closer to each domain's own specialist than the other specialist is when forced out-of-domain. Combined-tokenizer validation on the two visible NER benchmarks gives F1=0.4695 on domain 1 and F1=0.9611 on domain 2, in the same range as tokenizer 1 (0.4931 at its tuned vocab_size=4000) and tokenizer 2 (0.9623) on their own domains. Together this supports the expectation that tokenizer 3 should perform reasonably on a genuinely unseen domain. Caveat: this is validation across the two domains available for training, not proof about the actual hidden domain, which could differ stylistically from both. The one observed weakness is that tokenizer 3's fixed 32-slot bigram budget skews toward domain-1-style bigrams, since domain 1 contributes about four times more training lines than domain 2.